

# Anabel Yong

ay180501@gmail.com — Google Scholar: Link — Website: <https://anabelyong.github.io/>

## Summary

MSc in Computational Statistics & Machine Learning Graduate from UCL with interests in Gaussian Processes, Bayesian Optimisation, and kernel methods. Interested in machine learning for drug discovery and designing generative models for AI4Science.

## Publications

- **Yong, A.**, Tripp, A., Hosseini-Gerami, L., Paige, B. (2025). Bayesian Optimization for Molecules should be Pareto-Aware. (Accepted as Poster, NeurIPS 2025 AI for Science Workshop.) Link: arXiv

## Education

**MSc Comp. Statistics & Machine Learning, UCL (Distinction, Dean's List)** Sept 2023 - Sept 2024  
**Thesis Topic (87%):** *"Multi-Objective Bayesian Optimization with Independent Tanimoto Kernel Gaussian Processes for Diverse Pareto Front Exploration"*

**Modules:** Probabilistic & Unsupervised Learning (Gatsby PhD module) (91%), Machine Learning Seminar (Marc Deisenroth- DeepMind)(76%), Numerical Optimization (80%), Statistical Natural Language Processing

**BSc Maths & Biology, University of Edinburgh (First Class Honours)** Oct 2019 - June 2023

**Thesis Topic (80%):** *"Evaluation of different False discovery rate approaches in Large-scale Proteomics Data"*

**Modules:** Bioinformatics (95%), Machine Learning (80%), Numerical Methods for Differential Equations (85%), Linear Programming, Modelling & Solutions (76%), Mathematics for Natural Sciences (80%)

## Research Experience

- **Research Intern, National University of Singapore (PI: Jonathan Scarlett)** May 2025 - Present
  - Designing a batched Bayesian optimization algorithm to promote structural diversity in small molecule optimization.
- **Machine Learning Research Intern, IgnotaLabs.AI (PIs: Dr. Austin Tripp, Cambridge MLG Group & Brooks Paige, UCL)** June 2024 – Sept 2024
  - Developed a multi-objective Bayesian optimization algorithm using Tanimoto/MinMax Kernels with exact Gaussian Processes for small molecule optimization.
  - Benchmarked fixed weight scalarization(EI) versus standard Expected Hypervolume Improvement(EHVI) for molecular design.
- **Research Intern, Kustatscher Lab, University of Edinburgh** Oct 2022 - July 2023
  - Benchmarked a data pre-processing pipeline for proteomics, pgFDR), to enhance protein detection accuracy.
  - Aided in database curation of human proteins and microproteins of ProteomeHD2
- **Research Intern, ZiHeng Yang Lab, UCL** June 2022 - Sept 2022
  - Implemented Bayesian MCMC algorithms to model ABO blood type frequencies, contributing to population genetics.

## Skills

**Programming:** Python, R, MATLAB, git, Linux, Google Cloud Platform (GCP), slurm, L<sup>A</sup>T<sub>E</sub>X

**Machine Learning:** PyTorch, JAX, BoTorch, GPyTorch

**Bioinformatics/Cheminformatics:** RDKit, PyMOL, UniProt, MaxQuant

## Scholarships & Awards

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|-------------------------------------------------------------------|-----------------|
| Yayasan Sarawak Tun Taib Postgraduate MSc Scholarship 2023/2024   | Value: ≈£50,000 |
| EPSRC Undergraduate Research Award (University of Edinburgh) 2021 | Value: £2,500   |
| University of Edinburgh Biological Chemistry 2 Award 2020/2021    | NA              |

## References (available upon request)

- **Brooks Paige, Associate Professor in Machine Learning, University College London:** [b.paige@ucl.ac.uk](mailto:b.paige@ucl.ac.uk)
- **Jonathan Scarlett, Associate Professor in Dept. of Computer Science & Mathematics, National University of Singapore:** [scarlett@comp.nus.edu.sg](mailto:scarlett@comp.nus.edu.sg)
- **Layla Hosseini-Gerami, Chief Data Science Officer, Ignotalabs.ai:** [layla.gerami@ignotalabs.ai](mailto:layla.gerami@ignotalabs.ai)
- **Austin Tripp, Senior Machine Learning Research Scientist, Valence Labs:** [austin.james.tripp@gmail.com](mailto:austin.james.tripp@gmail.com)